



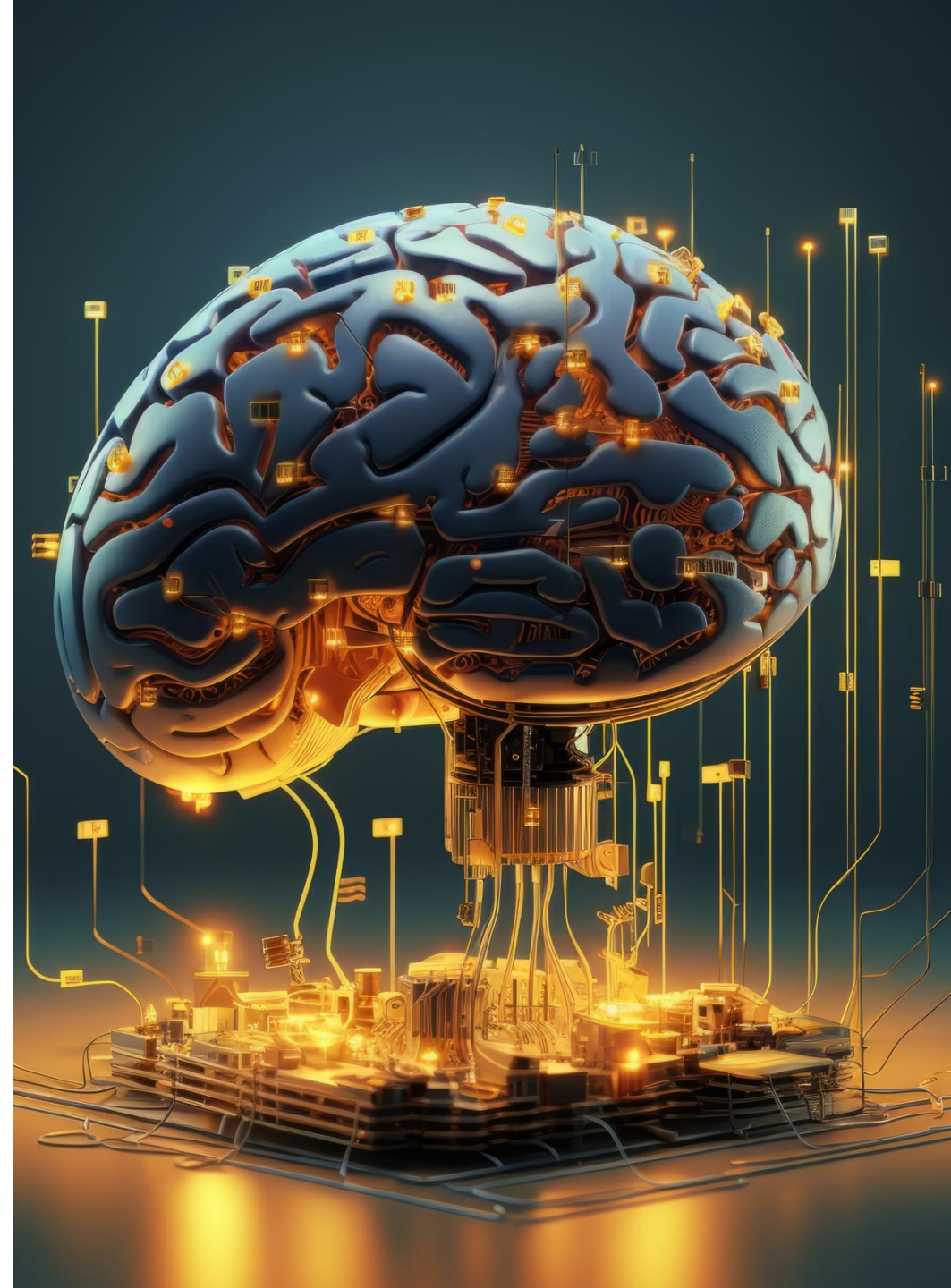
Intracranial Tumors in Adults An Overview

Khalid Shokor Mahmood



Epidemiology

- The annual incidence rate of all CNS tumors is about 25 per 100,000
- 30% are malignant and 70% are non-malignant
- An estimated 1 million Americans are living with a primary brain tumor
- 60% are females
- Unlike other types of cancer, primary brain tumors are **not staged**. They are grouped according to the WHO Classification of Tumors of the CNS, which assigns a “**grade**” of 1-4 based on how aggressive the tumor is predicted to behave.



Presenting Features

- ❖ **Progressive neurologic deficit (68%):**
motor weakness (45%). (notice that stroke causes more rapid presentation)
- ❖ **Headache: in 54%. May occur with or without raised ICP. (with nausea and vomiting in 40%)**
- ❖ **Seizures in 26%**





Neurologic deficits associated with brain tumors

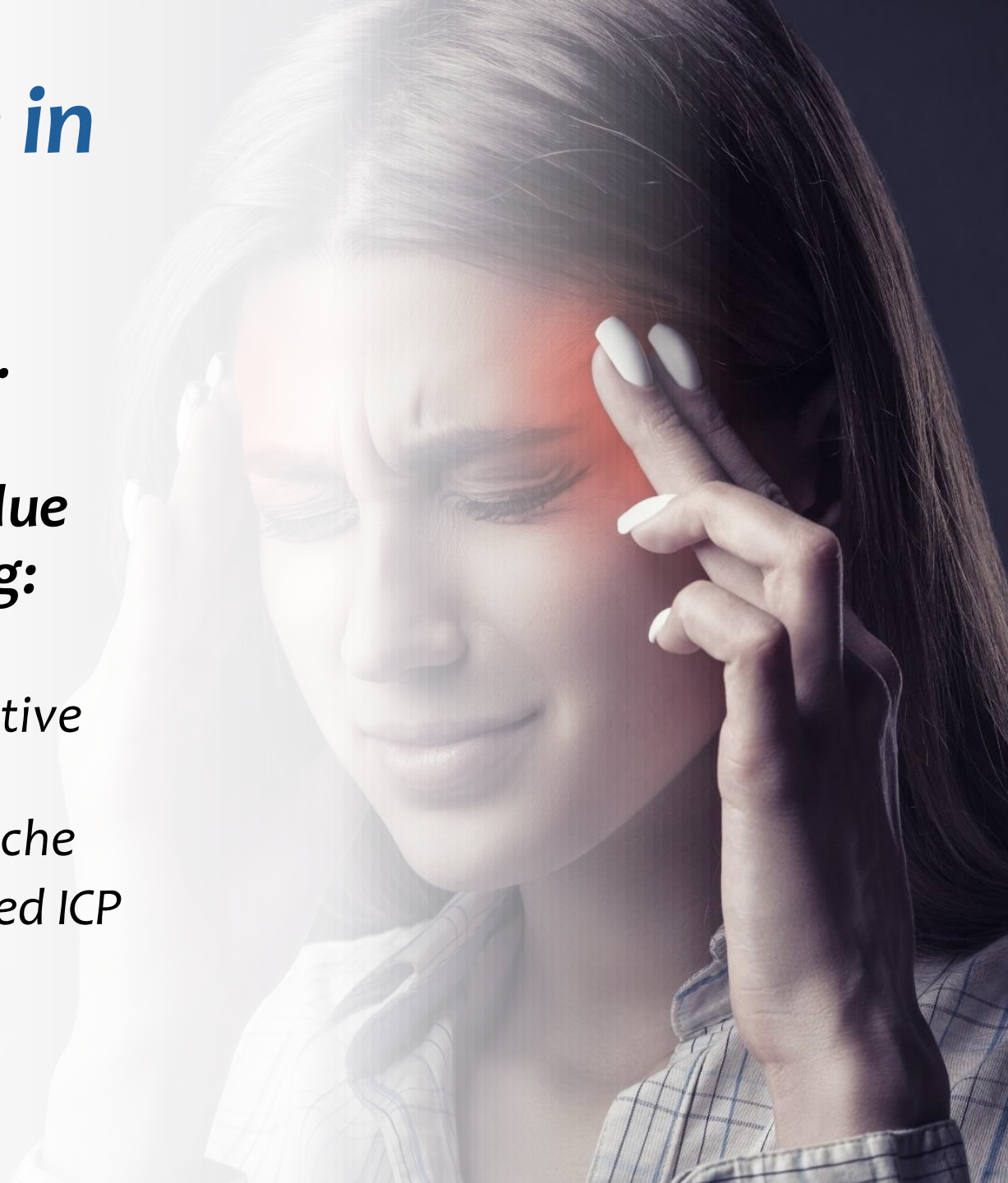
Brain tumors may produce progressive deficits related to the function of the involved brain.:

- 1. frontal lobe:** abulia, dementia, personality changes. Apraxia, hemiparesis or dysphasia (with dominant hemisphere involvement)
- 2. temporal lobe:** auditory or olfactory hallucinations, déjà vu, memory impairment.
Contralateral superior quadrantanopsia may be detected on visual field testing
- 3. parietal lobe:** contralateral motor or sensory impairment, homonymous hemianopsia.
Agnosias (with dominant hemisphere involvement) and apraxias
- 4. occipital lobe:** contralateral visual field deficits, alexia (especially with corpus callosum involvement with infiltrating tumors)
- 5. posterior fossa:** cranial nerve deficits, ataxia (truncal or appendicular)



Etiologies of Headache in Intracranial Tumors

- **The brain itself is not pain sensitive.**
- **Headache in brain tumors may be due to any combination of the following:**
 1. increased ICP
 2. invasion or compression of pain sensitive structures
 3. difficulty with vision can cause headache
 4. extreme hypertension due to increased ICP
 5. psychogenic



Supratentorial Tumors, C/F

- Features of raised ICP
- Progressive focal deficits
- Headache
- Seizures
- Mental status changes
- Symptoms suggestive of TIA
- Endocrine disturbances and pituitary apoplexy



Infratentorial Tumors, C/F

- Features of raised ICP
- Features of mass effect:
 - a) **Cerebellar hemisphere:** ataxia of extremities, dysmetria, intention tremor
 - b) **Cerebellar vermis:** broad based gait, truncal ataxia
 - c) **Brainstem:** multiple cranial nerve and long tract abnormalities nystagmus





Initial Evaluation

- **Imaging:** MRI $\pm$ contrast, pituitary study, imaging of entire neuraxis, MRV, MRA, MR spectroscopy (MRS).
- **Metastatic workup:** when metastasis is suspected
- **Lab:** Hormonal investigation in pituitary and sellar tumors, tumor markers in metastasis
- **Medications:**
 - **Steroids (Dexamethsones):** for lesions with significant or symptomatic edema or mass effect.
 - **Antiseizure medications (ASM):** indicated in some cases



Surgical Intervention vs Surveillance Mode

- Almost all tumors require surgical intervention.
 - Goals of surgery are usually:
 - cytoreduction (maximal safe resection) with the goal of gross total excision when feasible
 - biopsy for diagnostic, prognostic, and treatment planning purposes
-
- Surveillance with serial imaging is considered in situations with an extremely high confidence diagnosis of a slow growing benign lesion.
 - Use surveillance with caution with suspected low-grade glioma as some of these may be high-grade glioma or encephalitis.



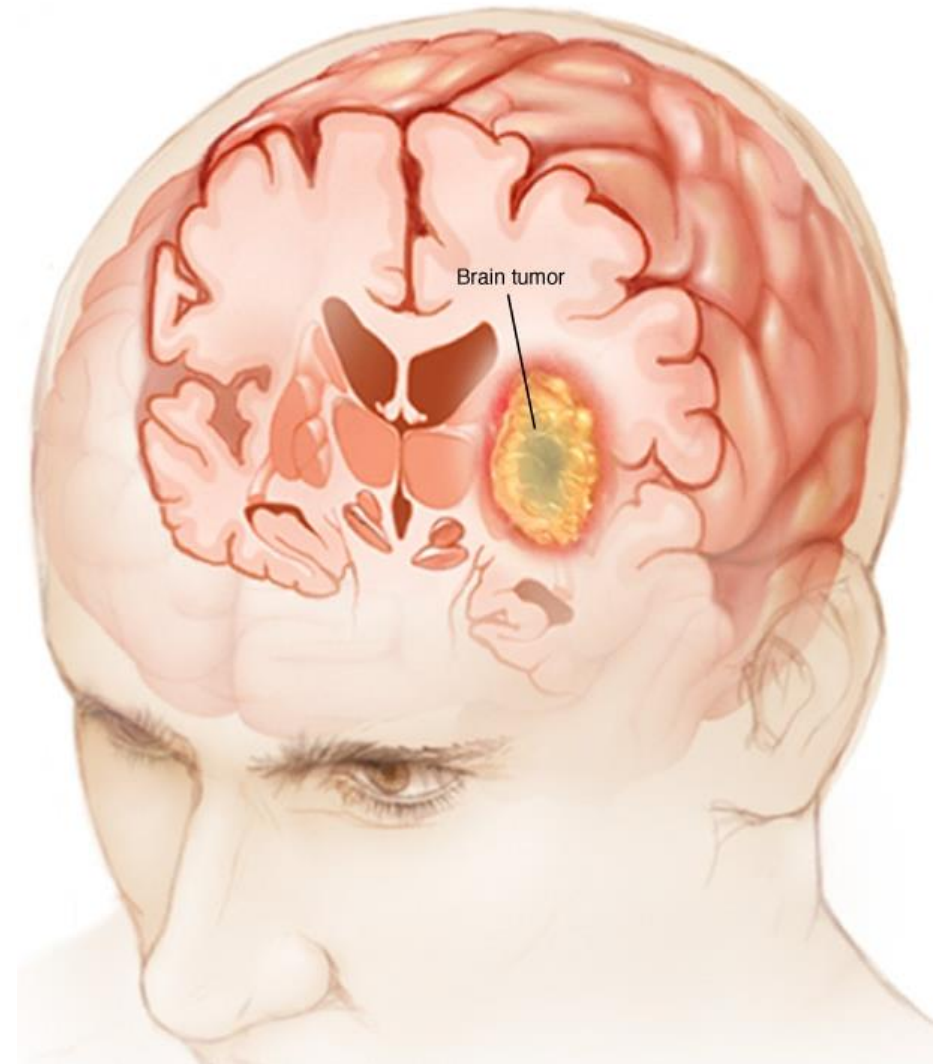


Adult Diffuse glioma

- These are the most common primary intra-axial brain tumors.
- Cell phones and gliomas?
- They spread by tracking in white matter, or less commonly through CSF pathways.
- **Types:**
 - **Astrocytoma, IDH-mutant (WHO grade 2, 3 or 4)**
 - **Oligodendroglioma, IDH mutant**
 - **Glioblastoma, IDH-wildtype (WHO grade 4)**

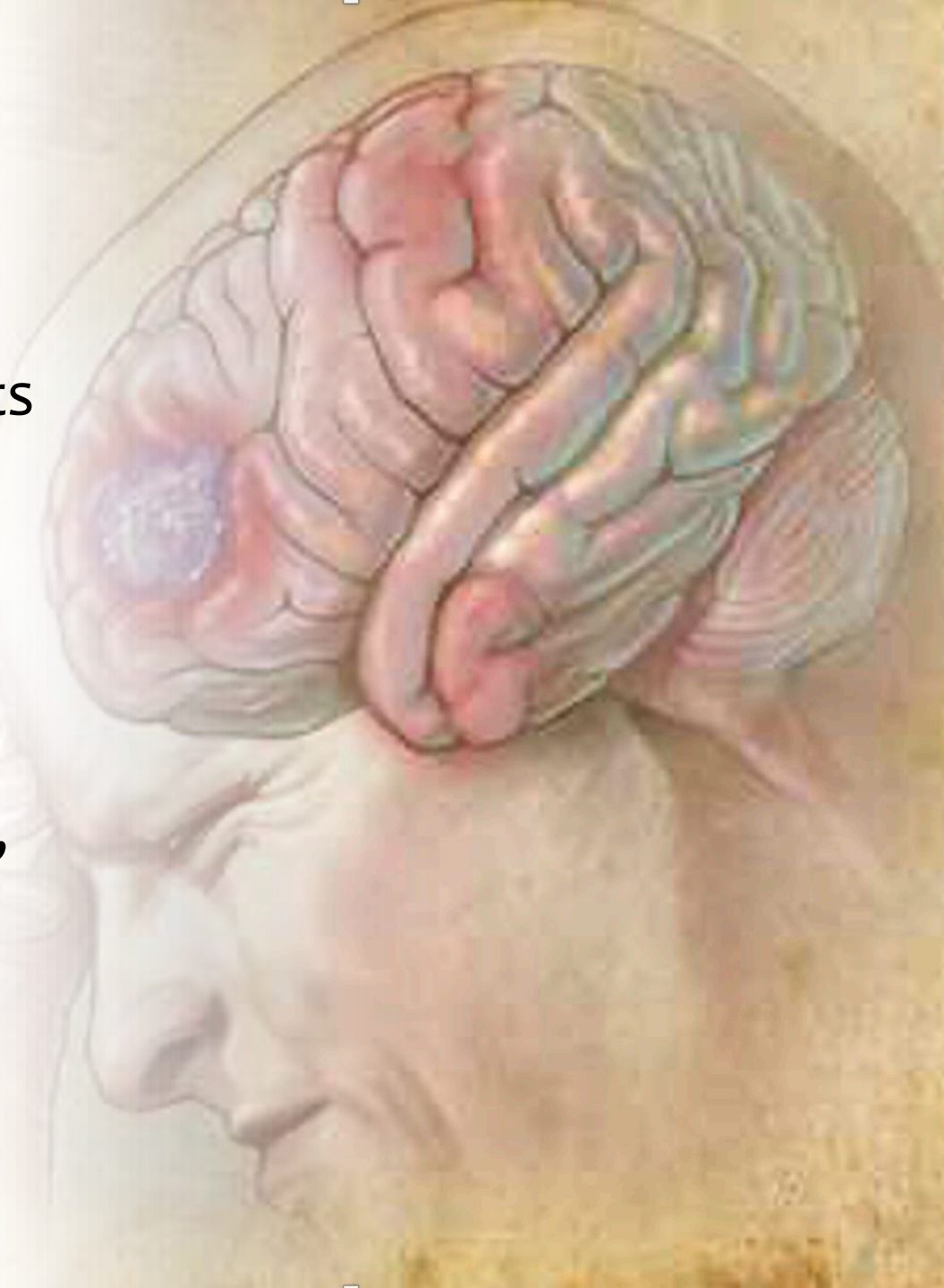
Astrocytoma, IDH-mutant (WHO grade 2, 3 or 4)

- *There is a predilection for the supratentorial compartment.*
- *Usually near or within the frontal lobes.*
- *Most present with seizures.*
- *Median survival is 11 years.*



Oligodendroglioma, IDH mutant

- Slow growing tumor, frequently presents with seizures
- Occur primarily in adults, frontal lobes > temporal lobe
- 2 grades: WHO 2 and 3
- Histology: classic features of “fried egg” cytoplasm. Calcifications are common



Glioblastoma, IDH-wildtype (WHO grade 4) (GBM)

- *The term “glioblastoma multiforme” is obsolete, although the abbreviation “GBM” persists.*
- This tumor arises de novo, without any identifiable lower grade precursor.
- **All GBM are now, by definition, IDH-wildtype.**
- Glioblastoma accounts for 14% of primary brain tumors, and is the most common adult malignant primary brain tumor (50% of them).
- ***It is also the most lethal***, with a median survival of 8 m, and five-year survival of 6.9%.
- male predominance of 1.6:1, and primarily affects older adults (mean age $\approx$ 62 years).



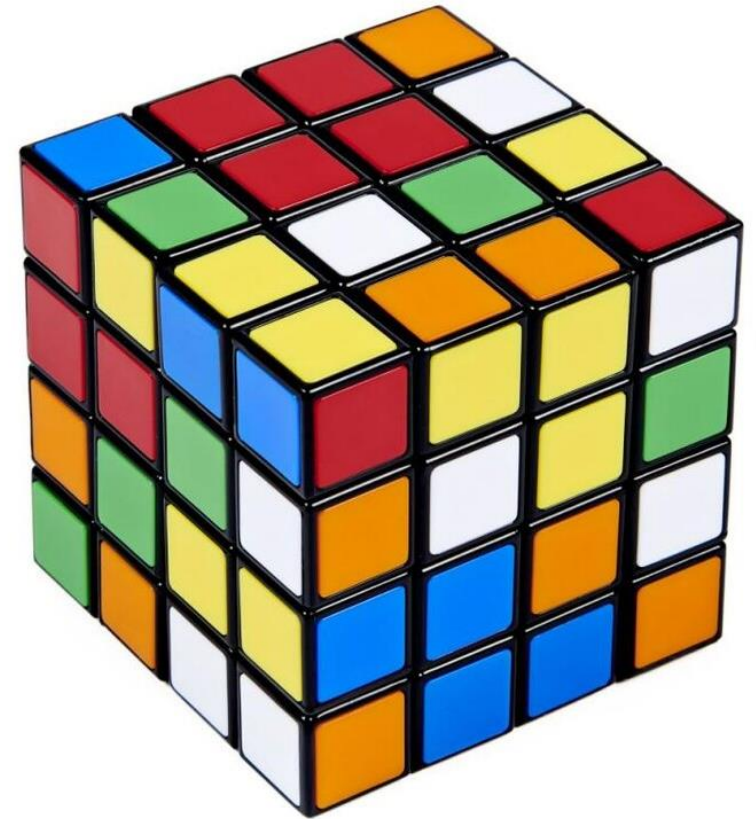
Surgical Intervention for Adult-Type Diffuse Gliomas

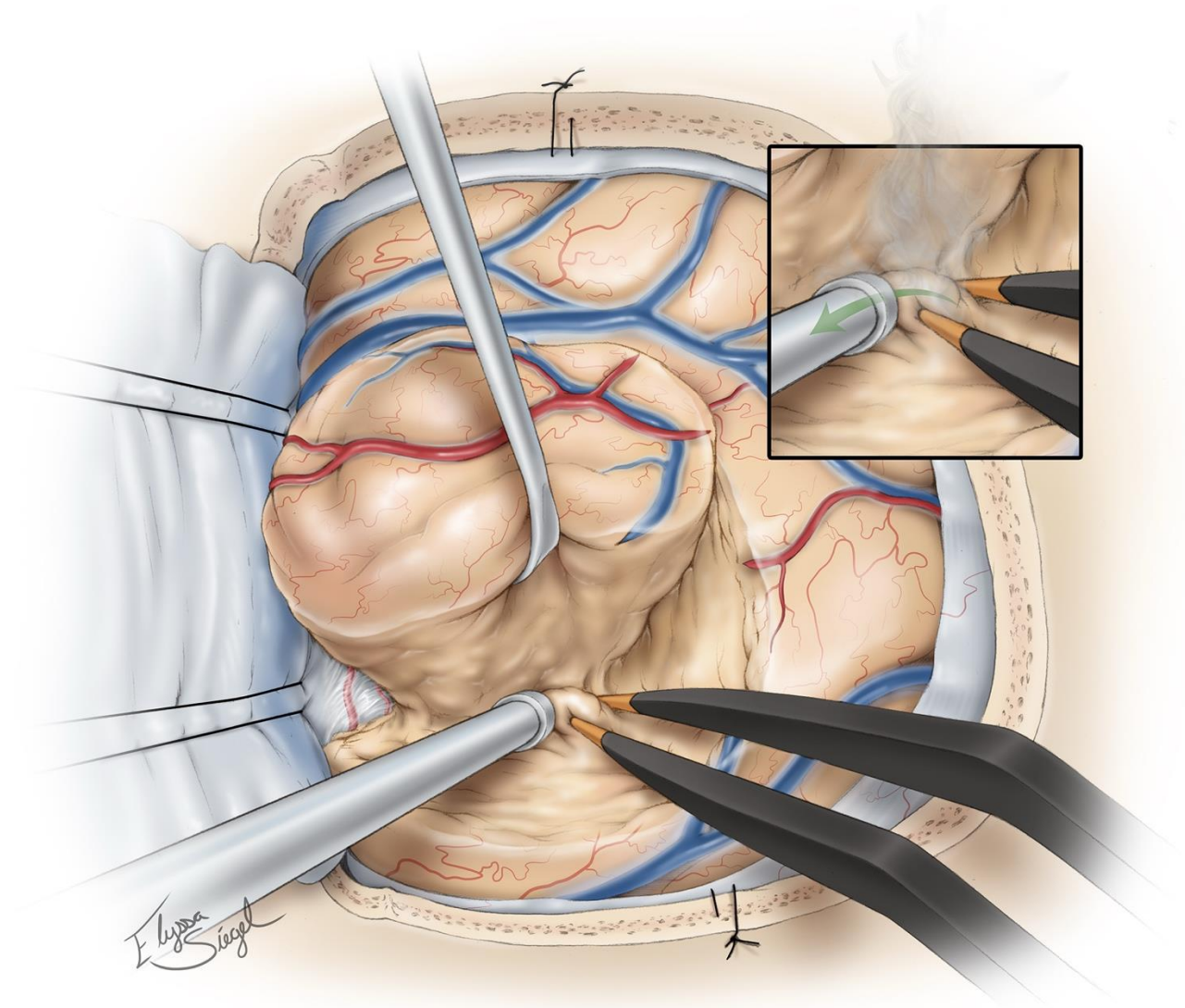
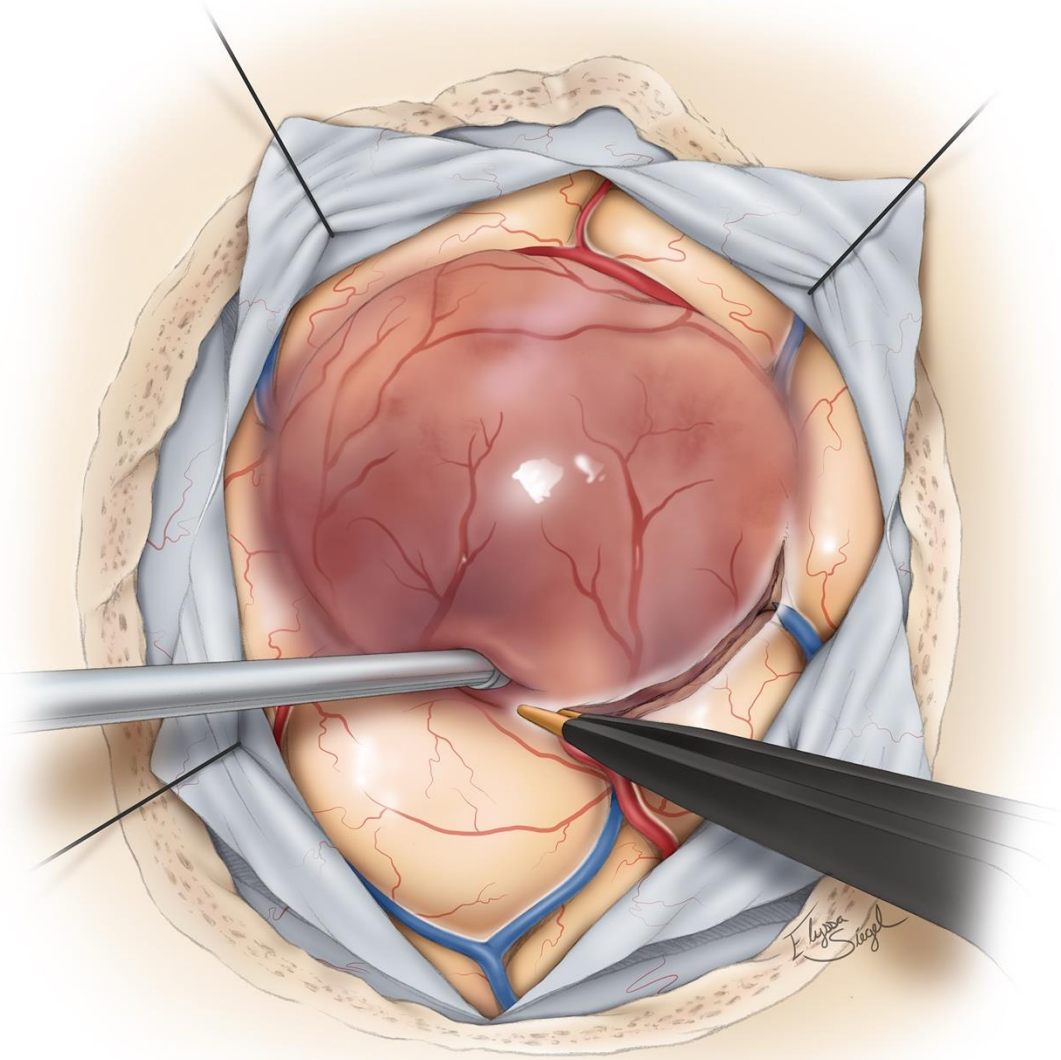
- Surgical intervention (cytoreduction or biopsy) is indicated for almost all tumors suspected of being diffuse gliomas.
- Surgery alone is never curative for these infiltrating lesions where there is no margin between brain and tumor.
- Aside from biopsy, the objective for cytoreduction is **maximal safe resection** (MSR) with the goal of gross total resection.



Maximal Safe Resection

- *MSR is an abstraction. In many cases, maximizing the resection increases the risk of surgery (decreasing the safety), and maximizing safety can mean compromising the extent of resection.*
- ***It is not possible to know the exact correct stopping point for MSR in every case.***





A few Words about Gross Total Resection

- It is acknowledged that there is a strong association between EOR and outcome, but the causality has not been established.
- We have to be satisfied with the proposition that maximizing the extent of resection likely increases overall survival (OS).

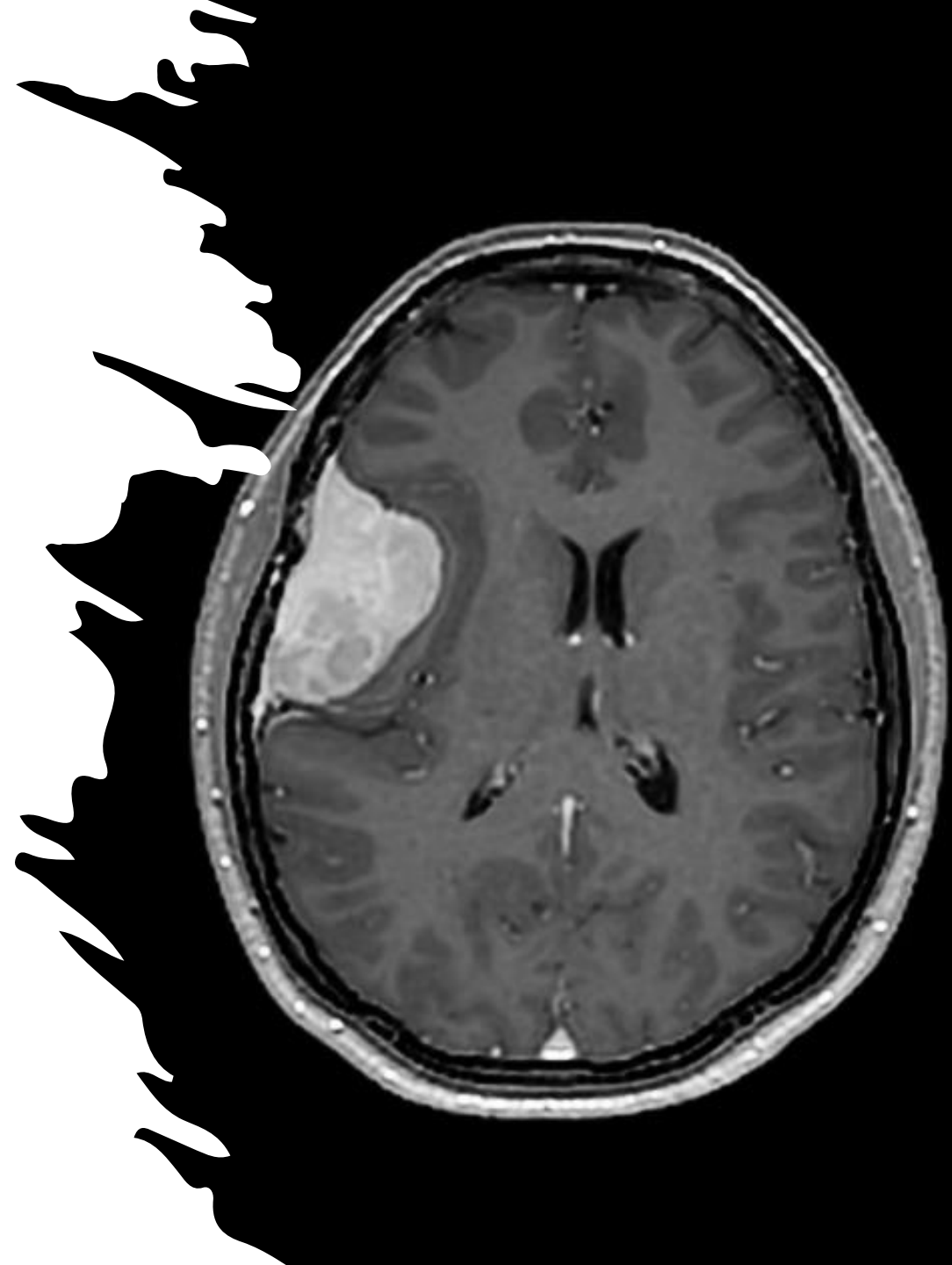
But keep in mind the following:

- ***Warning: the onset of a new postoperative motor or speech deficit reduces OS.***



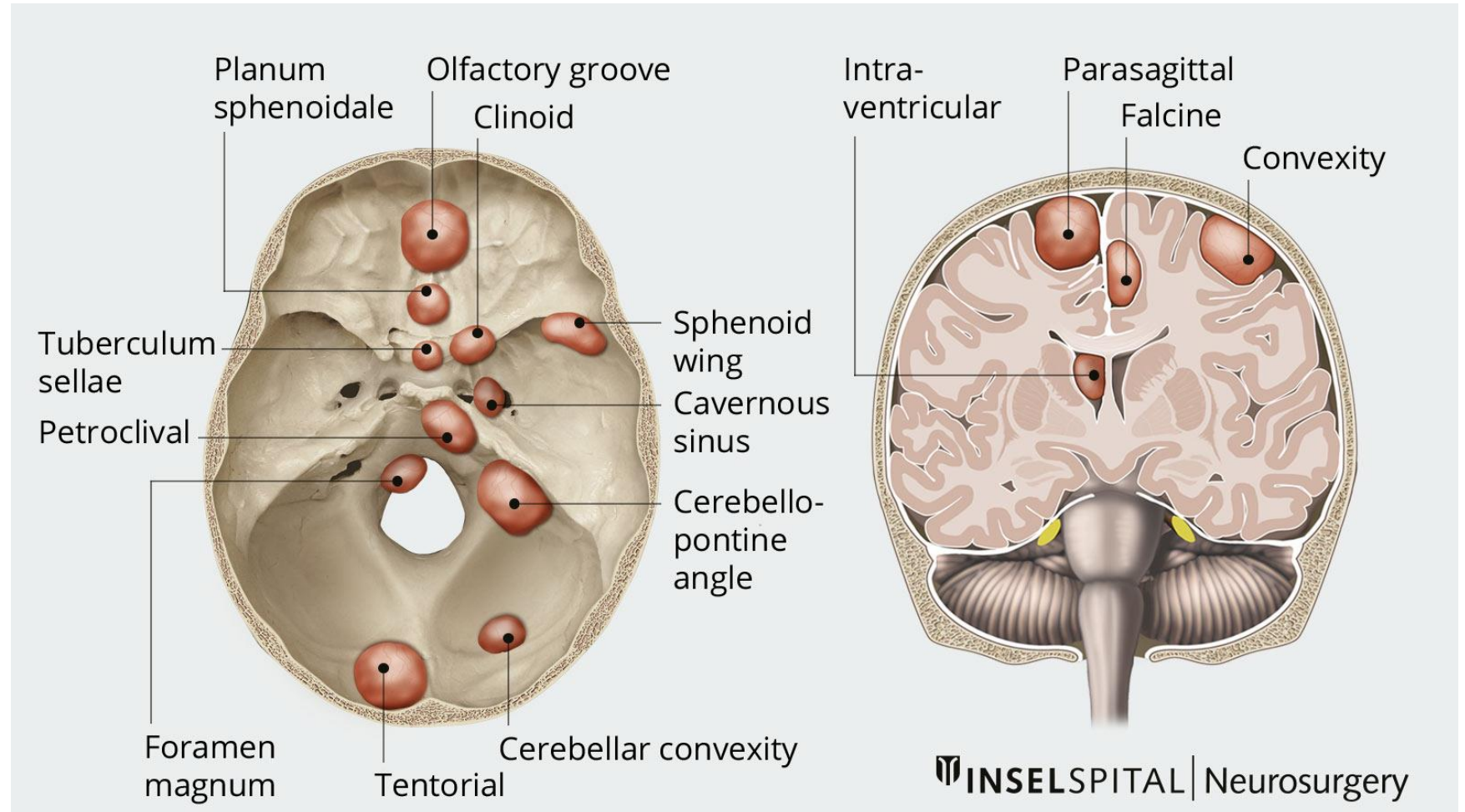
Meningiomas

- Most common primary intracranial tumors in adults, 36% of CNS tumors.
- They comprise 54.5% of non-malignant tumors.
- 10% are malignant and/or rapidly growing.
- They arise from arachnoid (not dura).
- 32% of incidentally discovered meningiomas do not grow over 3 years follow-up.
- Most (but not all) are cured if completely removed, which is not always possible.
- Most located along falx, convexity, or sphenoid bone.



Locations of Adult Meningiomas

Location	%
Parasagittal	20.8
Convexity	15.2
Tuberculum sellae	12.8
Sphenoidal ridge	11.9
Olfactory groove	9.8
Falx	8.0
Intraventricular	4.2
Tentorial	3.6
Middle fossa	3.0
Orbital	1.2
Spinal	1.2
Intrasylvian	0.3
Extracalvarial	0.3
Multiple	0.9





Epidemiology

Risk factors

- ionizing radiation
- hormonal influence
- neurofibromatosis

Genetics

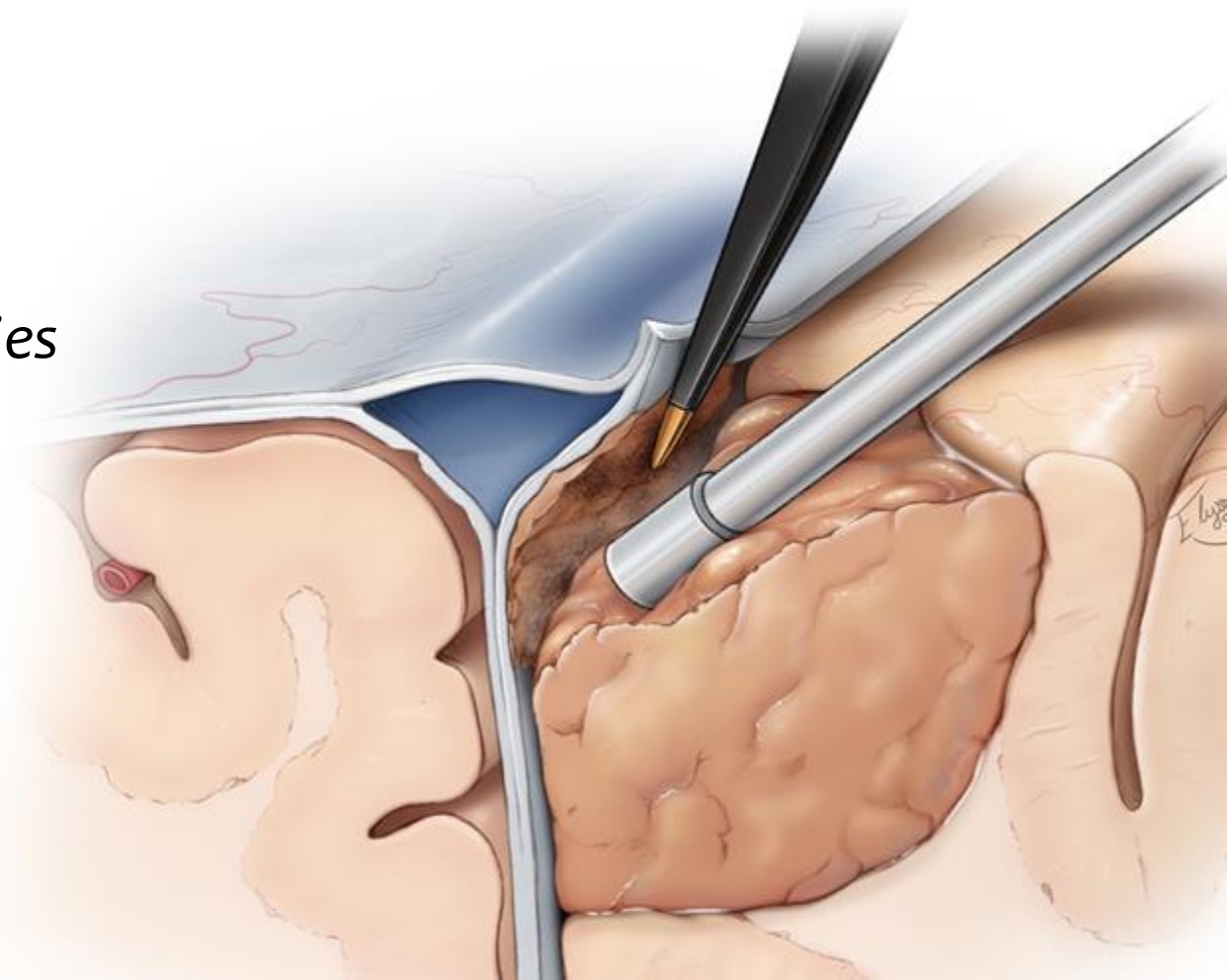
- 50-60% of WHO grade 1 meningiomas have mutations of the NF2 tumor suppressor gene on chromosome 22q12.2

General epidemiology

- 3% of autopsies on patients > 60 y of age reveal a meningioma
- Female:male ratio is 2.3:1

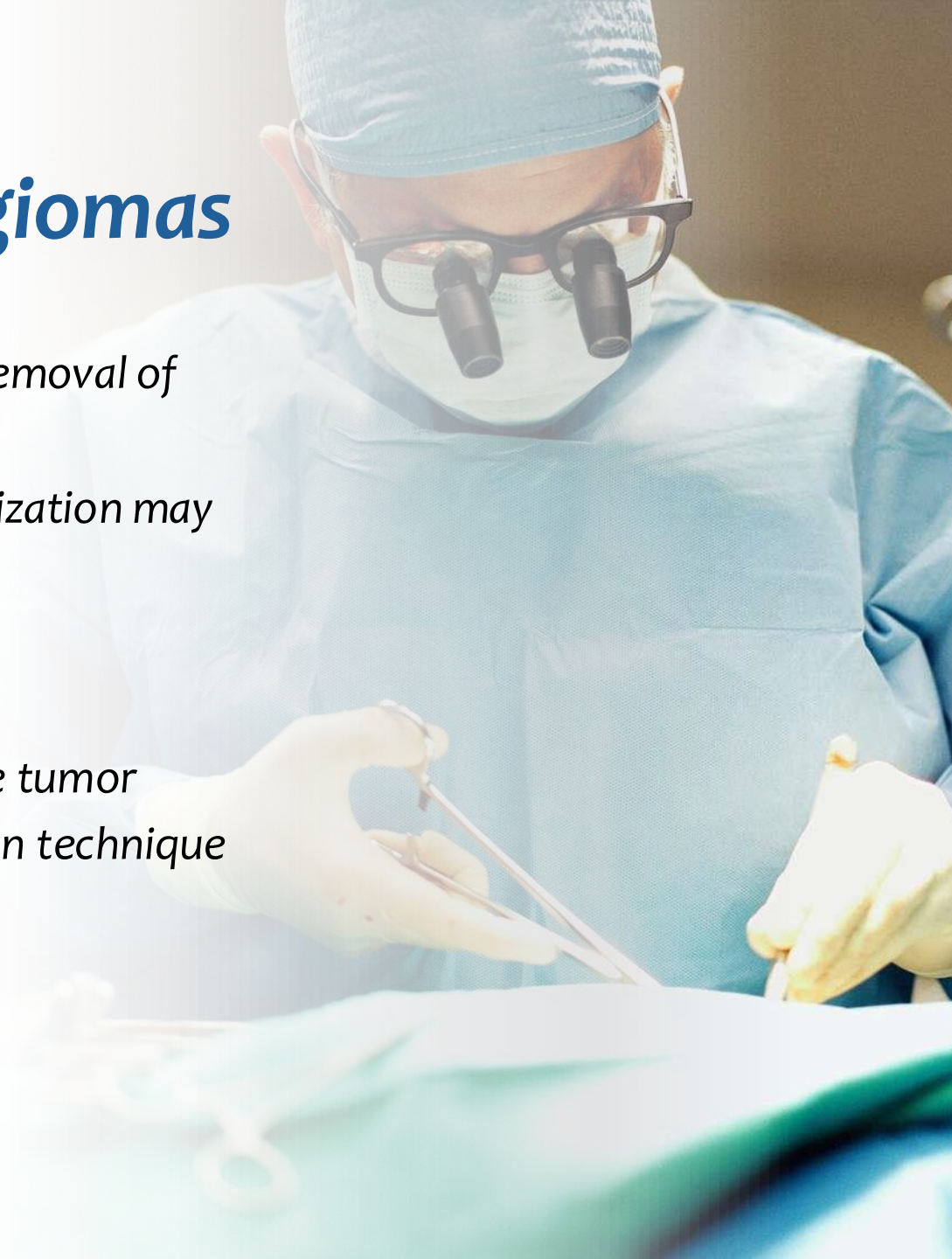
Indications for Treatment

- symptoms that cannot be satisfactorily controlled medically (e.g., refractory seizures)
- more than minimal edema on imaging
- significant growth on serial imaging studies
- Proposed criteria for “**significant growth**”:
 - 2-5mm increase in diameter over 1 year
 - 1 cc increase in volume in 1 year
 - 15% increase in volume over 1 year



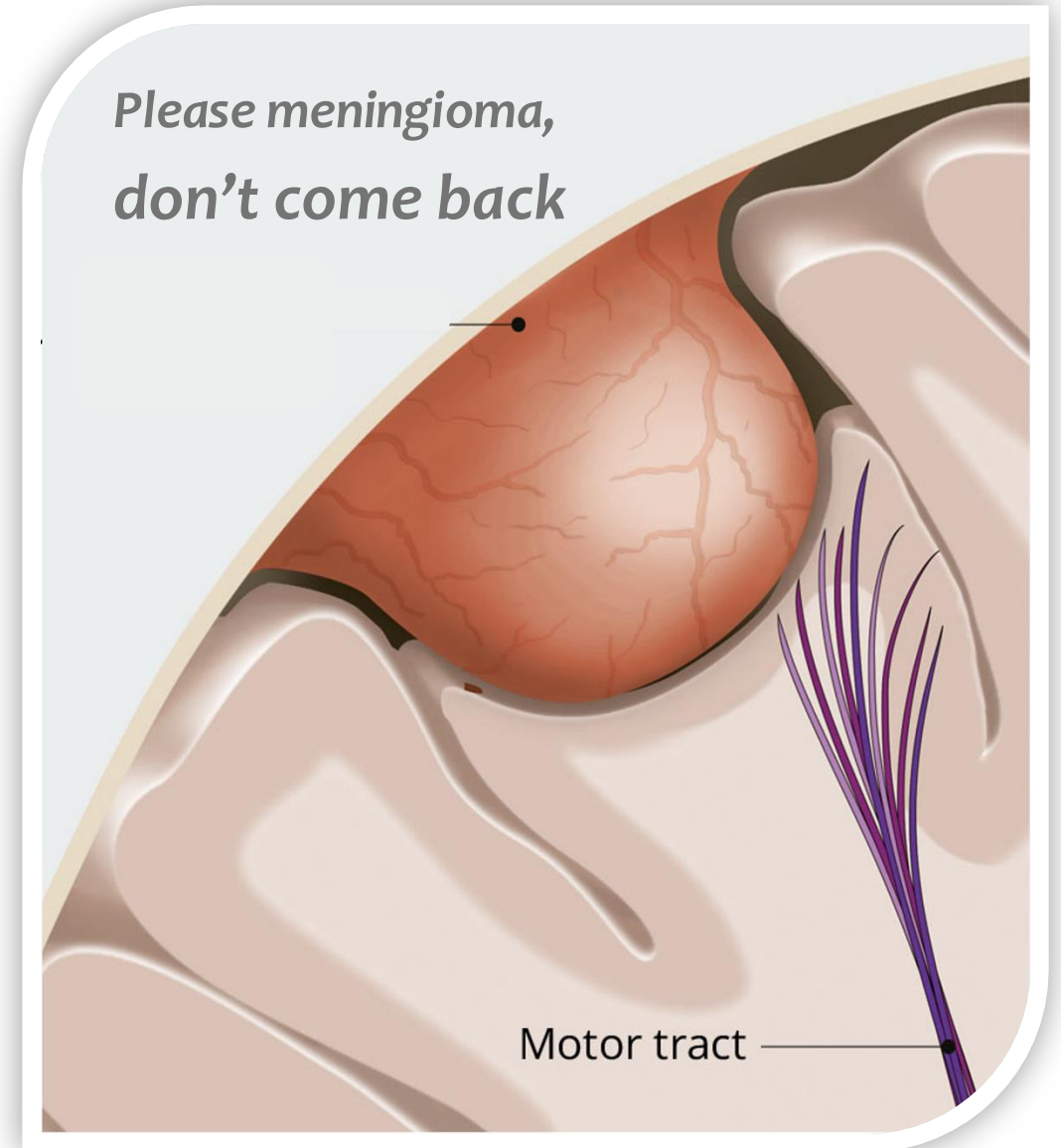
Surgical Treatment for Meningiomas

- Surgical goals: complete resection when possible, with removal of involved meninges and bone when feasible.
- Meningiomas are often very bloody. Preoperative embolization may be helpful for specific tumors.
- **General principles of meningioma surgery:**
 1. early interruption of the arterial blood supply to the tumor
 2. to avoid excessive retraction on the brain, a common technique consists of
 - a) internal decompression
 - b) followed by collapsing the tumor capsule
 3. removal of attached bone and dura when possible



Recurrence after Surgery

- The most powerful histopathologic predictor of recurrence is **WHO grade**:
 - grade 1: 7-25%
 - grade 2: 29-52%
 - grade 3: 50-94%
- The **extent of surgical removal** is the most important factor in the prevention of post-op recurrence.



Cerebral Metastases

- Brain metastases are the most common brain tumor seen clinically. Five times more common than primary brain tumors
- 25% of patients with cancer develop cerebral mets.
- In patients with no Ca history, a cerebral met was the presenting symptom in 15%
- With the onset of neurologic symptoms, 70% are multiple on MRI
- With solitary brain lesions in a patient with a history of cancer, biopsy should almost always be done since 11% of these lesions will not be mets
- Although median survival with maximal treatment is only 8 months (similar to GBM), long-term survivors do occur



Sources of cerebral mets in adults (Autopsy data)

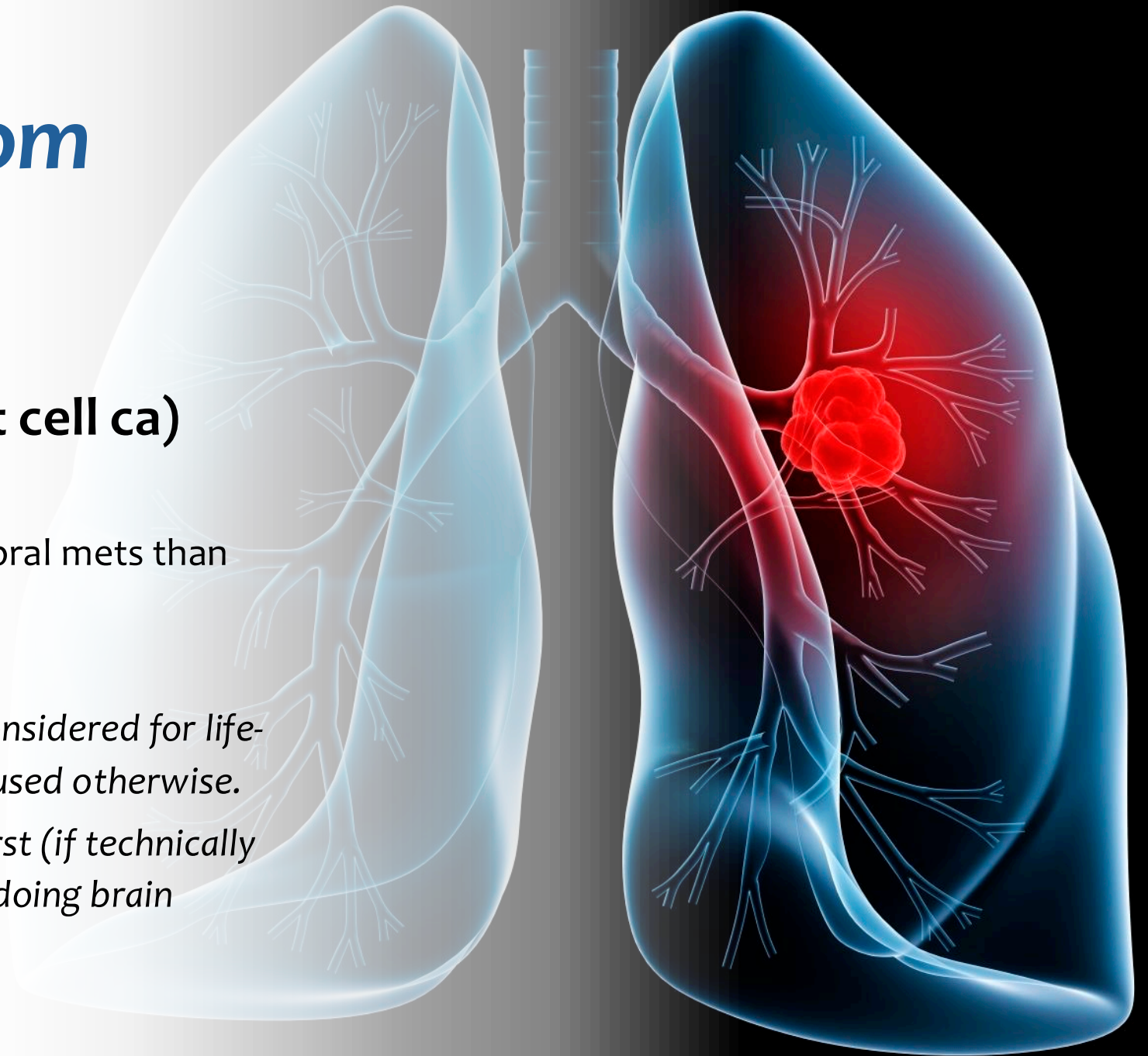
Primary	%
Lung	44
Breast	10
Renal	7
Colorectal	6
Melanoma	3
Undetermined	10

Autopsy incidence of cerebral mets for given primary cancers

Primary	%
Lung	21
Breast	9
Melanoma	40
Lymphoma	1
Colon	5
Gastric	0
Pancreatic	2
Kidney	21
Prostate	0
Testis	46
Cervix	5
Ovary	5
Osteosarcoma	10
Neuroblastoma	5
Head and neck	6

Cerebral Mets from Lung Cancer

- **Small-cell lung cancer (Oat cell ca)**
 - Median survival: 6-10 months
 - it is more likely to produce cerebral mets than other lung ca
 - Very radiosensitive
 - **Brain mets: surgical resection** considered for life-threatening large lesions, XRT is used otherwise.
 - Lung lesion should be biopsied first (if technically feasible) to rule out SCLC before doing brain surgery.



Evaluation of Cerebral Mets

- ***Imaging***
- ***Lumbar puncture***
 - Relatively contraindicated when there is a cerebral mass
- ***Metastatic work-up with suspected brain metastases***
 - CT TAP
 - Bone scan
 - Mammogram in women
 - PSA in men
 - PET scan



Management of Cerebral Mets

- With optimal treatment, median survival of patients with cerebral mets is still only $\approx 26-32$ weeks, therefore management is mostly palliative.
- **Management decisions:**
 - Prognostication
 - ***This is critical since many treatment decisions depend on overall prognosis.***
 - The specific tumor type, length of time since diagnosis, etc. are not as important prognostically as the Karnofsky Performance Scale (KPS) score.





Surgical Management for Solitary Mets

Indications favoring surgical excision:

- primary disease quiescent
- lesion accessible
- lesion is symptomatic or life-threatening
- primary tumor known to be relatively radioresistant
- for recurrent SCLC following XRT
- diagnosis unknown

Surgical Management for Multiple Mets

- much worse survival than solitary lesions.
- usually treated with XRT without surgery

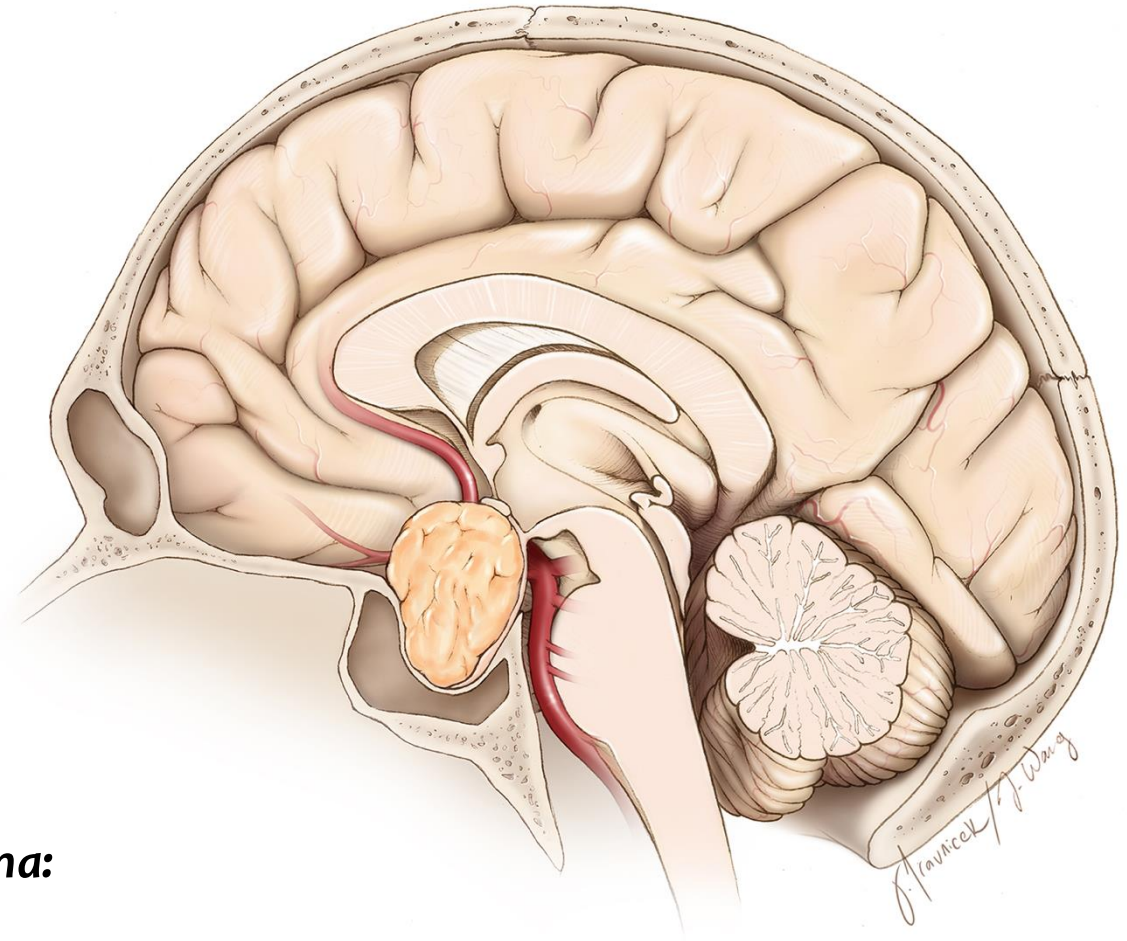
Indications favoring surgical excision:

1. *one particular and accessible lesion is clearly symptomatic and/or life threatening*
2. *multiple lesions that can all be completely removed*
3. *no diagnosis (e.g., no identifiable primary), consider biopsy*



Pituitary Tumors

- These account for about 10% of all intracranial tumors.
- Classified as the following:
 - **According the size:**
 - microadenomas (<1 cm)
 - macroadenomas (>1 cm) being either confined to the sella or with extrasellar extension
 - **According to the hormone secreted by the adenoma:**
 - Prolactinomas (40%)
 - Growth hormone adenomas (20%)
 - Non-functioning adenomas (20%)
 - ACTH adenomas (15%)





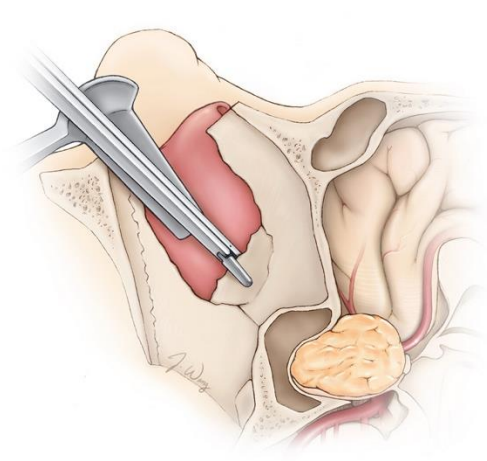
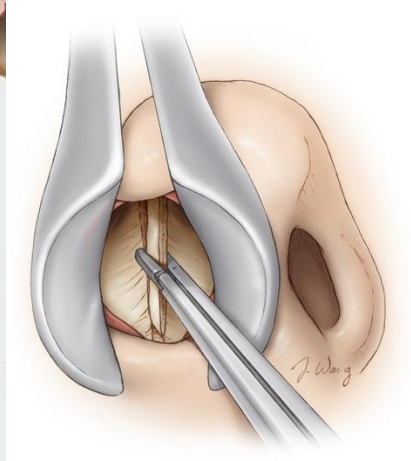
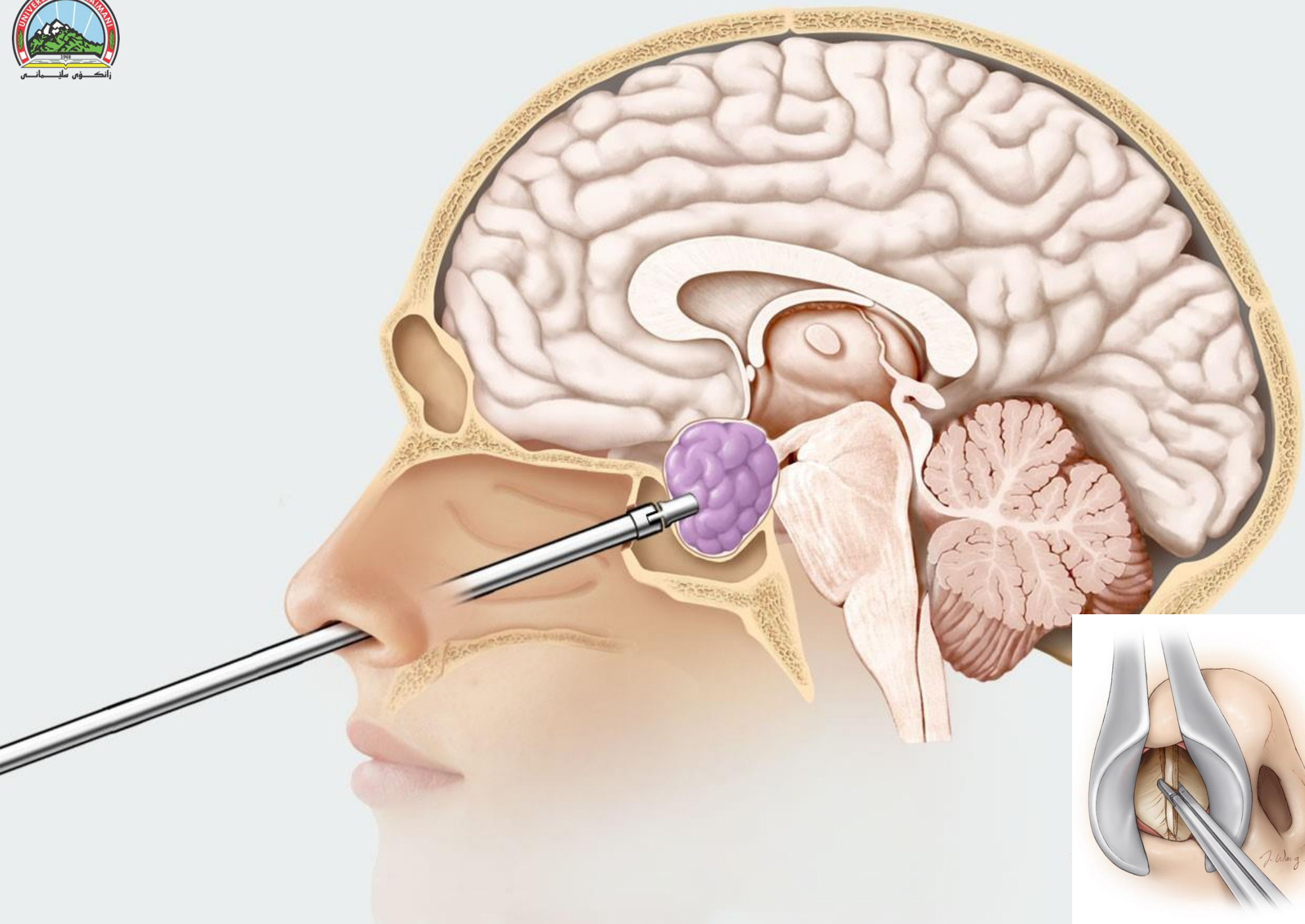
Clinical Features

- **Mass effect**
- **Endocrine dysfunction:**
 - *Hypopituitarism*
 - Prolactinomas
 - Growth hormone-secreting pituitary adenomas: acromegaly, or gigantism
 - *Cushing's disease* is due to ACTH producing pituitary adenomas
- **Pituitary apoplexy** results in the sudden onset of headache, visual loss, ophthalmoplegia and possibly altered conscious level. It is caused by *hemorrhagic infarction of a pituitary tumor*.



Treatment

- The aim of treatment of pituitary tumors is to alleviate mass effect, restore or replace normal endocrine function and prevent recurrence. Management requires close cooperation between neurosurgeon and endocrinologist.
 - **Conservative:** Prolactinomas should be initially treated medically with dopamine agonists such as cabergoline or bromocryptine. Growth hormone-secreting tumors may be amenable to medical treatment with somatostatin analogues such as octreotide or dopamine agonists.
 - **Surgical:** Done by **trans-sphenoidal** surgery either with an operating microscope or endoscope assisted. Large tumors with suprasellar extension may need to be managed by **craniotomy** as well.
- Replacement of the pituitary hormones is usually needed postoperatively.
- Radiotherapy is used for subtotal resection and for persistent hypersecretion of pituitary hormones.





Pediatric Brain Tumors

- Among all childhood cancers, brain tumors are the second only to leukemias in incidence (20%), and are the most common solid pediatric tumor, comprising 40–50% of all tumors.
- The common pediatric brain tumors (age < 18 years) are:
 - gliomas and glioneural tumors (45%)
 - embryonal tumors (12.3%) (like medulloblastoma)
 - pituitary tumors (6.2%)
 - nerve sheath tumors (4.2%)
 - craniopharyngiomas (3.8%)



Infratentorial vs. supratentorial tumor location

- *60% of pediatric brain tumors are infratentorial, and these are about equally divided among:*
 - *brainstem gliomas*
 - *cerebellar astrocytomas*
 - *medulloblastomas*

Thank You

